

Course Description

You'll gain hands-on experience in material testing, alloy composition, and microstructural analysis, and learn how to select and apply materials for real industrial purposes.

- Explain the relationship between structure, properties, and performance of metals and alloys
- Describe the phases in a metal alloy system using a phase diagram
- Apply heat treatment processes to modify metal properties for engineering applications
- Conduct lab tests to measure hardness, tensile strength, and ductility
- Interpret failure mechanisms — corrosion and fatigue — from properties and usage conditions
- Assess metal/alloy suitability for engineering scenarios: performance, durability, cost

NOTES

Assignment	Count	Value Each	Total Points
Attendance	15	1	15
Metallography Portfolio	1	50	50
Unit Quizzes	5	5	25
Final Exam	1	10	10
Total			100

Must be returned by 7:00 PM or the score is automatically zero.

The Metallography Portfolio (50 pts) is graded internally across all 15 weekly modules plus a holistic end-of-semester review — see the Metallography Lab slide for details.

NOTES

Course Timeline

Wk	Date	Topic	Wk	Date	Topic
1	Aug 24	Syllabus, PPE / Unit 1 begins	10	Oct 26	Unit 4 begins
2	Aug 31	Unit 1 (cont.) — Unit 1 Quiz	11	Nov 2	Unit 4 (cont.)
3	Sep 7	NO CLASS	12	Nov 9	Unit 4 (cont.)
4	Sep 14	Unit 2 begins	13	Nov 16	Unit 4 (cont.) — Unit 4 Quiz
5	Sep 21	Unit 2 (cont.)	14	Nov 23	Unit 5 begins
6	Sep 28	Unit 2 (cont.) — Unit 2 Quiz	15	Nov 30	Unit 5 (cont.)
7	Oct 5	Unit 3 begins	16	Dec 7	Unit 5 (cont.) — Unit 5 Quiz
8	Oct 12	Unit 3 (cont.)	17	Dec 16	FINAL EXAM
9	Oct 19	Unit 3 (cont.) — Unit 3 Quiz			

Tentative schedule — subject to change. Updates will be announced by the instructor.

SLIDE 6 *Course Timeline*

NOTES

Generative AI Policy

Academic Integrity

Participation Verification

Title IX

Leeward CC prohibits sex- and gender-based discrimination. Contact the Title IX Coordinator at leetix@hawaii.edu for resources and support.

NOTES

Specimen Schedule — What You'll Examine Each Week

Wk	Topic	Alloys Examined
1	Atomic Structure	1018 steel (new)
2	Mech. Properties	1018, 1045 steel (new: 1045)
3	NO CLASS	—
4	Carbon & Alloy Steels	1018, 1045, 1075 steel (new: 1075)
5	Stainless Steel	304, 316, 17-4PH SS (new)
6	Nonferrous Metals	6061 Al, 260 brass (new)
7	Mech. Testing	1018, 1045 steel (revisit)
8	NDT Methods	1018 steel, 360 brass, Ti-6Al-4V (new: 360, Ti)
9	Quality & Defects	1018 steel (revisit)

Wk	Topic	Alloys Examined
10	Wrought Aluminum	6061, 2024 Al (new: 2024)
11	Aluminum Forming	6063, 5052 Al (new)
12	Machining Processes	1018, 1045 steel, 6061 Al (revisit)
13	Copper Alloys	110, 182 Cu (new)
14	Brass Alloys	260, 360, 464 brass (new: 464)
15	Precip. Hardening	17-4PH SS, 2024 Al, 182 Cu (revisit)
16	Capstone & Synthesis	— (synthesis)

Each week's metallography module draws from the shared 16-specimen kit. "Revisit" means an alloy you've already examined, now viewed through a new lens.

SLIDE 10 Specimen Schedule — What You'll Examine Each Week

NOTES

